

Crash Course in Cyber

Marshall Nelson - 9/1/2026





Topics

- 1. What is “Cybersecurity”?**
- 2. Careers, Discipline, and Training Resources**
 - Operations
 - Software Development
 - Forensics and Incident Response
 - Threat Intelligence
 - Offensive Security
 - Policy / Legal
 - Research
- 3. Where to Start Today**

What Cybersecurity is NOT



What “Cybersecurity” Actually is.

Cybersecurity is a wide range of disciplines and skills with the primary goal of ensuring Confidentiality, Integrity, and Availability of digital information.



Careers, Disciplines, and Training



AUBURN

Operations

Cybersecurity Operations consists of the roles that keep systems and networks secure on a day-to-day basis.

Roles & Skills:

- Security Operations Center (SOC) Analyst
- Information Technology / Infrastructure
- Security Engineering (Network, SIEM, Tooling, Cloud, etc.)
- Detection Engineering

Operations: Training Resources

TryHackMe – Virtual SOC, Active Directory

Competitions – Hivestorm, SECCDC, Cyberforce, etc.

Homelabs – Infrastructure, SIEM Deployment, etc.

Certifications:

- Comp TIA Security+ (Udemy Courses are good)
- AWS Cloud Certs (Training by Amazon)
- Microsoft Entra ID

Software Development

Software Developers with a knowledge of cybersecurity.

Roles:

- Application Security (AppSec)
- Cybersecurity Tool/Platform Development

Training Resources:

Personal Projects - using encryption, secure dev practices, etc.

Competitions - NCL and PicoCTF (SQL Injection, Web Template Injection, Binary Exploitation, AppSec Analysis)

Digital Forensics and Incident Response (DFIR)

DFIR Specialists are responsible for what happens after a cybersecurity Incident.

Roles:

Digital Forensics Analyst (FBI, LEO, GOV)

Incident Response Analyst (Large Corps, Cyber Firms, Gov)

DFIR Training Resources

Competitions:

- NCL
- PicoCTF
- Jeopardy Style (Stego, File Recovery, Network log analysis)

Certifications (More Advanced):

- CompTIA CySA+
- EC-Council Certified Incident Handler (CIH)
- ISC2 Threat Handling Foundations Certificate

Threat Intelligence

Threat Intelligence is an advanced category of cybersecurity primarily concerning the prediction and detection of attacks from known high-level threats. This is typically a mid/late career role.

Roles & Skills:

Detection Engineering

Open-Source Intelligence (OSINT)

Strong Technical Knowledge

Offensive Security

Offensive security is where the popular culture idea of cybersecurity lives.

Roles:

Penetration Testing

Vulnerability Assessments

Exploit Research / Development
(AKA "Hackers")



OffSec Training Resources

Competitions:

- NCL, PicoCTF - Web Exploitation, Binary Exploitation, etc.
- Collegiate Penetration Testing Competition (CPTC)

Training:

TryHackMe

HackTheBox

OffSec (Ask James and Matthew)

Policy / Legal

These are the experts who make sure we don't break laws! Technical skills are only a portion of what these jobs require.

Roles & Skills:

Compliance Specialist

Compliance Auditor

Rules Of Engagement (RoE) Law

Research

Researchers exist both in industry and academia. They are the primary drivers for improvements and new technologies in the field.

Skills:

Academic Writing

Technical Knowledge (Core Specialization, can change)

High Self-Motivation

If interested, reach out to professors! Some professors are looking for undergraduates.

Where to Start!

Practice, Practice, Practice

1. PicoGym
2. HackTheBox
3. TryHackMe
4. Competition Trainings
5. Youtube Videos



AUBURN
UNIVERSITY